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FINAL EXAMINATION COVER PAGE

College of Natural and Social Science
 Department of Physics & Statistics
 Final Examination for the Course General Physics (Phys1001)

Abel Tessema

ETS 0027/14

A1

Examination Date: August 05, 2022

Maximum score: 50%

Time Allowed: 3:00 Hours.

Tick the box, where it applies you.

Freshman

Applied Science

☐

Engineering

☒

Regular:

Undergraduate

☒

Postgraduate

☐

Continuing Education

Undergraduate

☐

Postgraduate

☐

Course instructor(s) name(s) Haileyesus

General Directions

- Write your name (including your grandfather's name) and ID No. both on the question paper and answer sheet (if any).
- The examination booklet constitutes 4 type(s) of question(s), including: **Multiple Choice, Fill in the blank Spaces, Short Answer and Workout.**
- Please read the instructions carefully and attempt all the questions.
- Make sure that all the parts are included in your booklet, which has 8 page(s) including the cover page and answer sheet page(s), if any.
- It is strictly forbidden to bring mobile phones into the examination room, and unless permitted, no other electronic devices and materials are allowed.
- Use your own **SCIENTIFIC CALCULATOR**, Sharing calculator is not allowed!
- Don't use pencil or red pen.
- Don't detach the answer sheet, if any, from the question paper.
- Cheating automatically nullifies your total scores.

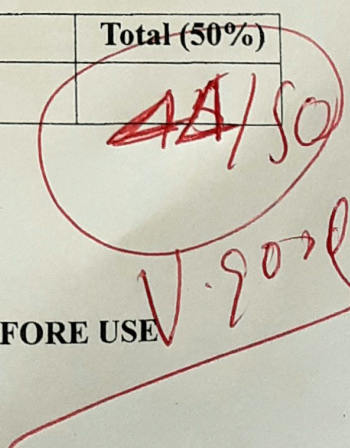
DO NOT START UNTIL YOU ARE TOLD TO DO SO!

For Instructor Use Only

Part I (7 %)	Part II (10 %)	Part III (8%)	Part III (25 %)	Total (50%)
7	10	8	19	44/50

WISH YOU ALL THE BEST!!

PLEASE MAKE SURE THAT THIS IS THE CORRECT ISSUE BEFORE USE


 V. 9028

Part I: Choose the correct answer from the given alternatives and write your answer in the space provided (7 %).

1. Let $\vec{A} = (2m)\hat{i} + (6m)\hat{j} - (3m)\hat{k}$ and $\vec{B} = (4m)\hat{i} + (2m)\hat{j} + (1m)\hat{k}$. Then $\vec{A} \cdot \vec{B} =$

A. $(8m)\hat{i} + (12m)\hat{j} - (3m)\hat{k}$

B. $(12m)\hat{i} - (14m)\hat{j} - (20m)\hat{k}$

C. $23 m^2$

D. $17 m^2$

$\vec{A} \cdot \vec{B} = a_x b_x + a_y b_y + a_z b_z$
 $= (2m)(4m) + (6m)(2m) + (-3m)(1m)$
 $= 8m^2 + 12m^2 - 3m^2 = 17m^2$

2. The inertia of a body tends to cause the body to:

A. speed up

B. slow down

C. resist any change in its motion

D. fall toward Earth

3. An elastic collision is one in which:

A. momentum is not conserved but kinetic energy is conserved

B. total mass is not conserved but momentum is conserved

C. kinetic energy and momentum are both conserved

D. momentum is conserved but kinetic energy is not conserved

4. A rock, which weighs 1400 N in air, has an apparent weight of 900 N when submerged in fresh water (998 kg/m^3). The volume of the rock is:

A. 0.14 m^3

B. 0.60 m^3

C. 0.90 m^3

D. $5.1 \times 10^{-2} \text{ m}^3$

$F_b = 1400 \text{ N} - 900 \text{ N} = 500 \text{ N}$

$= W_b = m_b g = \rho_b V_b g$

$V_b = \frac{F_b}{\rho_b g} = \frac{500}{998 \times 10} = \frac{50}{998} = 0.0501 \text{ m}^3 = 5.01 \times 10^{-2} \text{ m}^3$

5. A system undergoes an adiabatic process in which its internal energy increases by 20 J. Which of the following statements is true?

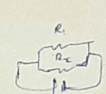
A. 20 J of work was done on the system

B. 20 J of work was done by the system

C. the system received 20 J of energy as heat

D. the system lost 20 J of energy as heat

Adiabatic $\Rightarrow Q = 0 \Rightarrow \Delta U = W$



$I_1 = \frac{V}{R_1}$

18.73

6. A particle with charge $2\text{-}\mu\text{C}$ is placed at the origin, an identical particle, with the same charge, is placed 2 m from the origin on the x axis, and a third identical particle, with the same charge, is placed 2 m from the origin on the y axis. The magnitude of the force on the particle at the origin is:

A. $9.0 \times 10^{-3} \text{ N}$

B. $6.4 \times 10^{-3} \text{ N}$

C. $1.3 \times 10^{-2} \text{ N}$

D. $1.8 \times 10^{-2} \text{ N}$

$q_3 = 2\mu\text{C}$

$F_{31} = \frac{k q_1 q_3}{r_{13}^2} = \frac{(9 \times 10^9)(2 \times 10^{-6})^2}{2^2}$
 $= \frac{(9 \times 10^9)(4 \times 10^{-12})}{4} = (9 \times 10^{-3} \text{ N})\hat{i}$

$F_{32} = \frac{k q_2 q_3}{r_{12}^2} = \frac{(9 \times 10^9)(4 \times 10^{-12})}{4}$
 $= (9 \times 10^{-3} \text{ N})\hat{j}$

$F_{\text{net}} = \sqrt{F_{31}^2 + F_{32}^2}$
 $= \sqrt{9^2 + 9^2}$
 $= 12.73 \times 10^{-3} \text{ N}$
 $= 1.273 \times 10^{-2} \text{ N}$

7. The current in the $5.0\text{-}\Omega$ resistor in the circuit shown is:

A. 0.42 A

B. 0.67 A

C. 1.5 A

D. 2.4 A

$R_{12} = \frac{R_1 R_2}{R_1 + R_2} = \frac{6 \times 12}{6 + 12} = \frac{6 \times 12}{18} = 4 \Omega$

$R_{123} = \frac{R_{12} R_3}{R_{12} + R_3} = \frac{4 \times 4}{4 + 4} = \frac{16}{8} = 2 \Omega$

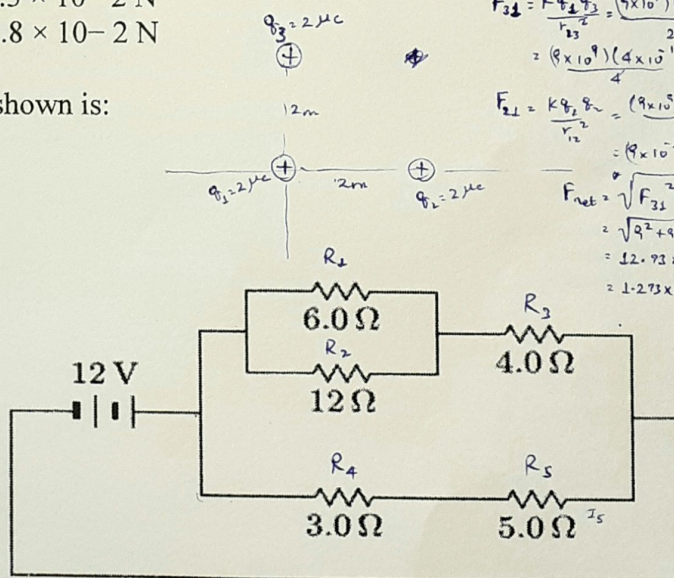
$R_{45} = R_4 + R_5 = 3 + 5 = 8 \Omega$

$R_{2345} = \frac{R_{123} R_{45}}{R_{123} + R_{45}} = \frac{2 \times 8}{2 + 8} = \frac{16}{10} = 1.6 \Omega$

$I_T = \frac{V_T}{R_T} = \frac{12}{1.6} = \frac{12 \times 5}{8} = \frac{45}{2} = 7.5 \text{ A}$

$I_{45} = \frac{V_T}{R_{45}} = \frac{12}{8} = \frac{3}{2} = 1.5 \text{ A}$

$I_5 = I_4 = I_{45} = 1.5 \text{ A}$



Part II. Fill in the Blank Spaces with the most simplified Answers. [10 pts.]

10

1. If an ideal gas absorbs $5 \times 10^3 \text{ J}$ of energy while doing $2 \times 10^3 \text{ J}$ of work on the environment during a constant pressure process, then the change in the internal energy of the gas will be $\boxed{3 \times 10^3 \text{ J}}$
2. If identical resistors of 8Ω are connected in parallel combination, their equivalent resistance is 2Ω . The number of resistors in the circuit are $\boxed{4}$
3. At one instant an electron ($q = -1.6 \times 10^{-19} \text{ C}$) is moving in the XY-plane and the components of its velocity is $V_x = 5 \times 10^5 \text{ m/s}$ & $V_y = 3 \times 10^5 \text{ m/s}$. If magnetic field of 0.8 T is in +X-direction, then the magnitude of magnetic force is $\boxed{3.84 \times 10^{-14} \text{ N}}$
4. If the temperature of a cube made of copper with initial volume 3m^3 , changes from 10°C to 15°C , its change in volume is $\boxed{7.65 \times 10^{-4} \text{ m}^3}$ (use: $\alpha = 17 \times 10^{-6} \text{ K}^{-1}$)
5. A hydraulic press has one piston of diameter 2cm & the other piston of diameter 8cm . What force must be applied to the smaller piston to obtain a force of 1600 N at the larger piston? $\boxed{100 \text{ N}}$

Answers for Part II

- 1) 3×10^3 ✓
- 2) 4 ✓
- 3) 3.84×10^{-14} ✓
- 4) 7.65×10^{-4} ✓
- 5) 100 ✓

$$5) \frac{F_1}{A_1} = \frac{F_2}{A_2} \Rightarrow F_1 = \frac{F_2 A_1}{A_2} = \frac{F_2 \pi d_1^2/4}{\pi d_2^2/4}$$

$$F_1 = F_2 \frac{d_1^2}{d_2^2} = \frac{(1600) \pi}{\pi 16} = 100 \text{ N}$$

	i	j
a	a_x	a_y
b	b_x	b_y

$$4) \Delta V = \gamma V_0 \Delta T = 3 \alpha V_0 \Delta T$$

$$= 3 (17 \times 10^{-6} \text{ K}^{-1}) (3 \text{ m}^3) (5 \text{ K})$$

$$= 765 \times 10^{-6} \text{ m}^3 = 7.65 \times 10^{-4} \text{ m}^3$$

$$1) Q = 5 \times 10^3 \text{ J}, W = -2 \times 10^3 \text{ J}$$

$$2) \Delta U = Q + W = 5 \times 10^3 - 2 \times 10^3 = 3 \times 10^3 \text{ J}$$

$$2) \frac{1}{R_T} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} \Rightarrow \frac{1}{R_T} = \frac{3}{R}$$

$$\frac{1}{R} = \frac{1}{R_T} \Rightarrow n = \frac{R}{R_T} = \frac{8}{2} = 4$$

$$3) q = -1.6 \times 10^{-19} \text{ C}, \vec{V} = (5 \times 10^5 \text{ m/s})\hat{i} + (3 \times 10^5 \text{ m/s})\hat{j}$$

$$\vec{B} = (0.8 \text{ T})\hat{i}, F = ?$$

$$\vec{F} = q(\vec{V} \times \vec{B}) = q(V_x \hat{i} + V_y \hat{j}) \times B_x \hat{i}$$

$$= (-1.6 \times 10^{-19}) (3 \times 10^5 \times 0.8) \hat{j}$$

$$= -1.6 \times 3 \times 0.8 \times 10^{-14} = -3.84 \times 10^{-14} \text{ N} \hat{j}$$

Part III. Give a short and brief answer for the following questions. (8 %)

8

1. State:

a) the Zeroth law of thermodynamics

"If an object 'A' is in thermal equilibrium with an object 'B', and if object 'B' is in thermal equilibrium with an object 'C', then it follows that object 'A' is in thermal equilibrium with object 'C'. (If $T_A = T_B$ and $T_B = T_C$, then $T_A = T_C$.)"

b) the first law of thermodynamics

"The change in internal energy of a system is equal to the sum of the heat flow into the system and the work done on the system." ($\Delta U = Q + W$)

2. State:

a) Faradays law of electromagnetic induction

"The size of the induced EMF in a circuit is directly proportional to the rate of change of the magnetic flux [and the number of turns for a coil]."

b) Lenz's law

"The direction of the induced EMF is such as to oppose the change that's causing it."

$$\mathcal{E} = -N \frac{\Delta \Phi}{\Delta t}$$

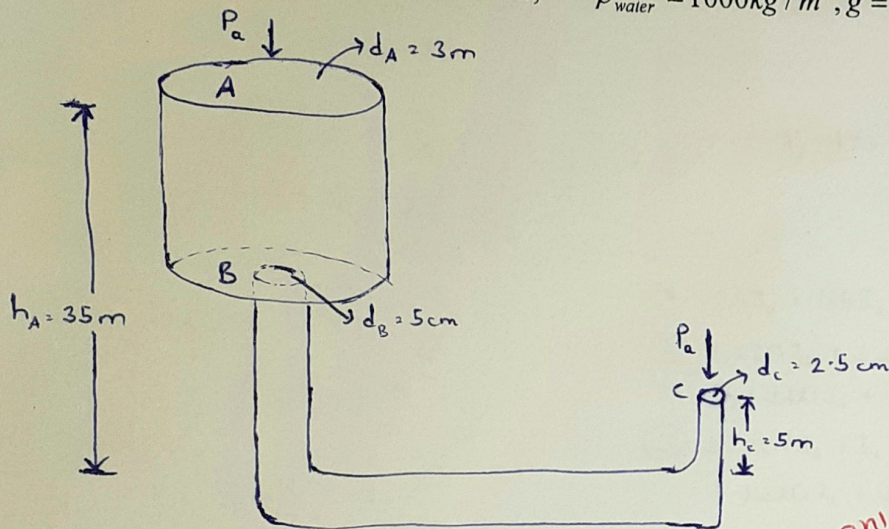
$$\mathcal{E} = -N \frac{\Delta \Phi}{\Delta t} \rightarrow \text{Faraday's Law}$$

↓
Lenz's Law

Part IV. Workout the Following Problems and show your steps clearly & Neatly. [25%]

1. A cylindrical water tower of diameter 3 m supplies water to a house. The level of water in the water tower is 35 m above the point where the water enters the house through a pipe that has an inside diameter 5 cm. The intake pipe delivers water at a maximum rate of $2 \times 10^{-3} \text{ m}^3 \cdot \text{s}^{-1}$. The pipe is connected to a narrower pipe leading to the second floor that has an inside diameter 2.5 cm.
- A) Calculate the pressure of the water in the narrow pipe at 5 m height. (3 pts.)
 B) Find the speed of the water in the narrow pipe. (3 pts.)

(Use: $P_a = 1 \text{ atm} = 101.3 \text{ KPa}$, $\rho_{\text{water}} = 1000 \text{ kg/m}^3$, $g = 10 \text{ m/s}^2$)



- A) Since the total pressure is constant throughout, $P_c = P_A$.

$$P_c = P_A + \rho g h_A = 101,300 \text{ pa} + (1000 \text{ kg/m}^3 \times 10 \text{ m/s}^2 \times 35 \text{ m})$$

$$= 101,300 \text{ pa} + 350,000 \text{ pa} = 451,300 \text{ pa}$$

$$= 451.3 \text{ kPa}$$

- B) $P_c = P_a + \rho g h_c + \frac{1}{2} \rho v_c^2$

$$\frac{1}{2} \rho v_c^2 = P_c - P_a - \rho g h_c = 451,300 \text{ pa} - 101,300 \text{ pa} - (1000 \text{ kg/m}^3 \times 10 \text{ m/s}^2 \times 5 \text{ m})$$

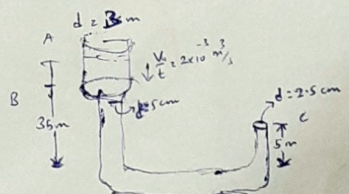
$$= 350,000 \text{ pa} - 50,000 \text{ pa} = 300,000 \text{ pa}$$

$$\rho v_c^2 = 600,000 \text{ pa}$$

$$v_c^2 = \frac{600,000 \text{ pa}}{1000 \text{ kg/m}^3} = 600 \text{ m}^2/\text{s}^2$$

$$v_c = \sqrt{600 \text{ m}^2/\text{s}^2} = 10\sqrt{6} \text{ m/s} = 24.5 \text{ m/s}$$

$$P_1 + \rho g h_1 + \frac{1}{2} \rho v_1^2 = P_2 + \rho g h_2 + \frac{1}{2} \rho v_2^2$$



$$\frac{\text{kg}}{\text{m}^3} \times \frac{\text{m}^2}{\text{s}^2} = \frac{\text{kg}}{\text{m}^3}$$

$$\sqrt{25 \times 24} = 5\sqrt{4 \times 6} = 10\sqrt{6}$$

$$24.5 \text{ m/s}$$

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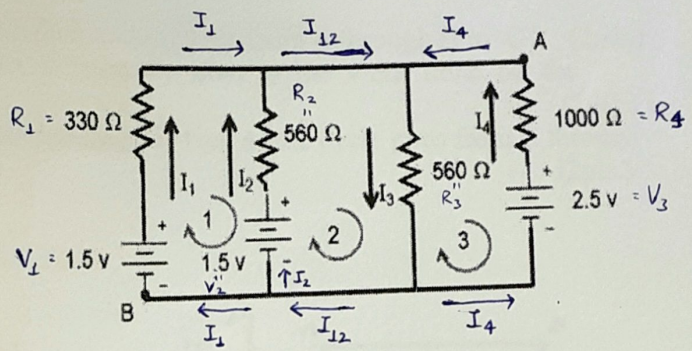
2. In the circuit shown below, use Kirchhoff's Law and find:
- The currents in each resistors. (2 pts.)
 - The voltage drops across the resistors (2 pts.)
 - The potential difference V_{AB} between points A and B (2pts.)

6

A

$$I_{12} = I_1 + I_2$$

$$I_3 = I_{12} + I_4 = I_1 + I_2 + I_4$$



Loop 1

$$V_1 - I_1 R_1 + I_2 R_2 - V_2 = 0$$

$$1.5 - 330 I_1 + 560 I_2 - 1.5 = 0$$

$$330 I_1 = 560 I_2$$

$$I_1 = \frac{56}{33} I_2$$

Loop 2

$$V_2 - I_2 R_2 - I_3 R_3 = 0$$

$$1.5 - 560 I_2 - 560 I_3 = 0$$

$$3 - 1,120 I_2 - 1,120 I_3 = 0$$

$$1,120 I_2 + 1,120 I_3 = 3$$

Loop 3

$$I_3 R_3 + I_4 R_4 - V_3 = 0$$

$$560 I_3 + 1000 I_4 - 2.5 = 0$$

$$1,120 I_3 + 2000 I_4 = 5$$

- $33 I_1 = 56 I_2 \Rightarrow 33 I_1 - 56 I_2 = 0$
- $1,120 I_2 + 1,120 (I_4 + I_2 + I_4) = 3$
 $\Rightarrow 1,120 I_2 + 2,240 I_4 = 3$
- $1,120 (I_1 + I_2 + I_4) + 2,000 I_4 = 5$
 $\Rightarrow 1,120 I_1 + 1,120 I_2 + 3,120 I_4 = 5$

$$\begin{cases} 33 I_1 - 56 I_2 + 0 I_4 = 0 \\ 1,120 I_1 + 2,240 I_2 + 1,120 I_4 = 3 \\ 1,120 I_1 + 1,120 I_2 + 3,120 I_4 = 5 \end{cases}$$

$$D = \begin{vmatrix} 33 & -56 & 0 \\ 1,120 & 2,240 & 1,120 \\ 1,120 & 1,120 & 3,120 \end{vmatrix} = 33(2,240 \times 3,120 - 1,120^2) + 56(1,120 \times 3,120 - 1,120^2) = 3.15 \times 10^8$$

$$D_x = \begin{vmatrix} 0 & -56 & 0 \\ 3 & 2,240 & 1,120 \\ 5 & 1,120 & 3,120 \end{vmatrix} = 56(3 \times 3,120 - 5 \times 1,120) = 2.1 \times 10^5$$

$$D_y = \begin{vmatrix} 33 & 0 & 0 \\ 1,120 & 3 & 1,120 \\ 1,120 & 5 & 3,120 \end{vmatrix} = 33(3 \times 3,120 - 5 \times 1,120) = 1.24 \times 10^5$$

$$D_z = \begin{vmatrix} 33 & -56 & 0 \\ 1,120 & 2,240 & 3 \\ 1,120 & 1,120 & 5 \end{vmatrix} = 33(2,240 \times 5 - 1,120 \times 3) + 56(1,120 \times 5 - 1,120 \times 3) = 3.84 \times 10^5$$

$$I_1 = \frac{D_x}{D} = \frac{2.1 \times 10^5}{3.15 \times 10^8} = 6.67 \times 10^{-4} \text{ A}$$

$$I_2 = \frac{D_y}{D} = \frac{1.24 \times 10^5}{3.15 \times 10^8} = 3.94 \times 10^{-4} \text{ A}$$

$$I_4 = \frac{D_z}{D} = \frac{3.84 \times 10^5}{3.15 \times 10^8} = 1.22 \times 10^{-3} \text{ A}$$

$$I_3 = I_1 + I_2 + I_4 = 2.28 \times 10^{-3} \text{ A}$$

C) $V_{AB} = V_B - V_A$
 $= V_3 - I_4 R_4$
 $= 2.5 - 1.22$
 $= 1.28 \text{ V}$

- B) For $R_1 \Rightarrow V = I_1 R_1 = 0.22 \text{ V}$
 For $R_2 \Rightarrow V = I_2 R_2 = 0.22 \text{ V}$
 For $R_3 \Rightarrow V = I_3 R_3 = 1.28 \text{ V}$
 For $R_4 \Rightarrow V = I_4 R_4 = 1.22 \text{ V}$

Abel Tesserra
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Handwritten calculations and a large red checkmark.

- 6
3. In the Fig below, the change in internal energy of a gas that is taken from A to C is $= +800 \text{ J}$. The work done on the gas along path ABC is $= -500 \text{ J}$.
- How much energy must be added to the system by heat as it goes from A through B to C? (2pts.)
 - If the pressure at point A is five times that of point C, what is the work done on the system in going from C to D? (2pts.)
 - What is the energy exchanged with the surroundings by heat as the cycle goes from C through D to A. (2pts.)

a) $\Delta U = Q + W$

$$Q = \Delta U - W = 800 \text{ J} - (-500 \text{ J})$$

$$= 1,300 \text{ J}$$

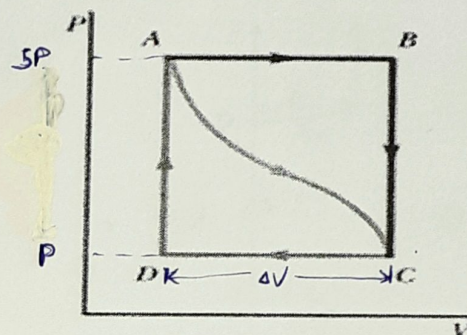
b) $W_{AB} = \text{Area under the graph}$

$$= 5P \Delta V = 500 \text{ J}$$

$$\Rightarrow P \Delta V = \frac{500 \text{ J}}{5} = 100 \text{ J}$$

$W_{CD} = \text{Area under the graph}$

$$= P \Delta V = 100 \text{ J}$$



$W =$

$$W_{AB} = 4P \Delta V$$

$$-500 = 5P \Delta V \Rightarrow P \Delta V = -\frac{500}{5} = -100$$

$$W_{CD} = P \Delta V$$

$$-800 = Q + 100$$

$$Q = -900$$

- c) Returning from 'C' to 'A', it loses all the internal energy it gained before. Thus, from 'C' through 'D' to 'A', $\Delta U = -800 \text{ J}$.

$$\Delta U = Q_{CD} + W_{CD} + Q_{DA} + W_{DA}$$

$$Q_{CD} + Q_{DA} = \Delta U - W_{CD} = -800 \text{ J} - 100 \text{ J} = -900 \text{ J}$$

Four identical charges, each having a charge of $+q = 4 \times 10^{-6} \text{ C}$, are fixed at the corners of a square of side $L = \sqrt{8} \text{ m}$.

- Find the electric field at a point P that lies at a distance $Z = 2 \text{ m}$ along the line perpendicular to the plane of the square and passing through the center of the square. (Fig. below) (3pts.)
- Find the net electric force on a fifth point charge $+q = 4 \times 10^{-6} \text{ C}$, at a point P that lies at $Z = 2 \text{ m}$ above the center of the square due to the other four charges. (2pts.)
- Calculate the potential energy due to the four charge's distribution located on the square. (2pts.)

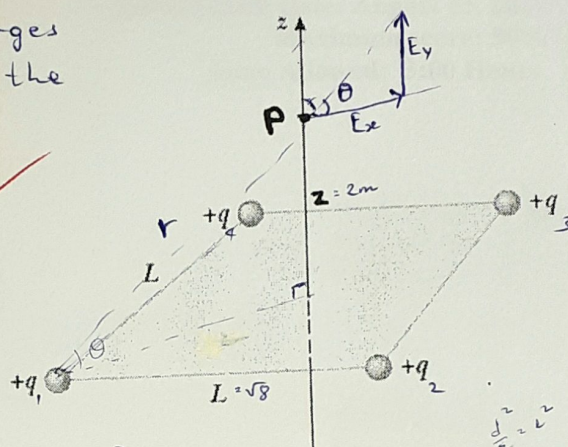
(use: $K = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$)

- A) $Z = 2 \text{ m}$, Let r be the distance from the charges to the point 'P', & d be the diagonal of the square plane.

$$A = \frac{d^2}{2} = L^2 \Rightarrow d^2 = 2L^2 = 2 \times 8 = 16 \text{ m}^2 \Rightarrow d = 4 \text{ m} \checkmark$$

$$d_{1/2} = \frac{d}{2} = \frac{4 \text{ m}}{2} = 2 \text{ m} \checkmark$$

$$r = \sqrt{Z^2 + d_{1/2}^2} = 2\sqrt{2} \text{ m} \checkmark$$



When adding up the electric fields, the net electric field will be only along the z axis, & upwards, since the rest of the components cancel out. The E_{net} will be four times the vertical component of the electric fields, since all four of them are equal.

$$E_{\text{net}} = \frac{4Kq \cdot \sin\theta}{r^2} = \frac{4(9 \times 10^9)(4 \times 10^{-6})}{8} \cdot \frac{\sqrt{2}}{2} = 1.8 \times 10^4 \text{ N/C} \times \frac{\sqrt{2}}{2} = 1.27 \times 10^4 \text{ N/C}$$

$$B) F_{\text{net}} = E_{\text{net}} \cdot q = (1.8 \times 10^4)(4 \times 10^{-6}) = 7.2 \times 10^{-2} \text{ N} \cdot \frac{\sqrt{2}}{2} = 5.09 \times 10^{-2} \text{ N}$$

$$C) U = K \left(\frac{q_1 q_2}{\sqrt{8}} + \frac{q_2 q_3}{\sqrt{8}} + \frac{q_3 q_4}{\sqrt{8}} + \frac{q_4 q_1}{\sqrt{8}} + \frac{q_1 q_3}{4} + \frac{q_2 q_4}{4} \right)$$

$$= (9 \times 10^9)(4 \times 10^{-6})^2 \left(\frac{4}{\sqrt{8}} + \frac{1}{2} \right) = (9 \times 10^9)(16 \times 10^{-12})(1.91)$$

$$= 2.76 \times 10^{-1} = 0.276 \text{ J}$$

Abel Tesserma

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